TOPOS MORPHISM G FROM THE LISSE-ETALE AND FLAT-FPPF SITES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Allowed standard prerequisites (editorial).

- Continuous and cocontinuous functors and shriek existence. Sites, Definitions sites-definition-continuous and sites-definition-cocontinuous; Sites, Lemmas sites-lemma-when-shriek and sites-lemma-back-and-forth; Modules on Sites, Lemmas sites-modules-lemma-g-shriek-adjoint and sites-modules-lemma-back-and-forth. All clauses.

Source reference. Original reference 00WV: sources/stacks-chow/tags.txt, source lines 1197–1197. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00XJ: sources/stacks-chow/tags.txt, source lines 1221–1221. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00XR: sources/stacks-chow/tags.txt, source lines 1229–1229. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00XT: sources/stacks-chow/tags.txt, source lines 1231–1231. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04BG: sources/stacks-chow/tags.txt, source lines 5651–5651. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 077I: sources/stacks-chow/tags.txt, source lines 9203–9203. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack.*

(1) *The inclusion functor $\mathcal{X}_{lisse,étale} \to \mathcal{X}_{étale}$ is fully faithful, continuous and cocontinuous. It follows that*
   (a) *there is a morphism of topoi*
   $$g : Sh(\mathcal{X}_{lisse,étale}) \longrightarrow Sh(\mathcal{X}_{étale})$$
   with g^{-1} given by restriction,
   (b) *the functor g^{-1} has a left adjoint $g_!^{Sh}$ on sheaves of sets,*
   (c) *the adjunction maps $g^{-1}g_* \to id$ and $id \to g^{-1}g_!^{Sh}$ are isomorphisms,*
   (d) *the functor g^{-1} has a left adjoint $g_!$ on abelian sheaves,*
   (e) *the adjunction map $id \to g^{-1}g_!$ is an isomorphism, and*

Date: stacks-project:stacks-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(f) *we have* $g^{-1}\mathcal{O}_{\mathcal{X}} = \mathcal{O}_{\mathcal{X}_{lisse,étale}}$ *hence* g *induces a flat morphism of ringed topoi such that* $g^{-1} = g^*$.

(2) *The inclusion functor* $\mathcal{X}_{flat,fppf} \to \mathcal{X}_{fppf}$ *is fully faithful, continuous and cocontinuous. It follows that*

(a) *there is a morphism of topoi*

$$g : Sh(\mathcal{X}_{flat,fppf}) \longrightarrow Sh(\mathcal{X}_{fppf})$$

with g^{-1} *given by restriction,*

(b) *the functor* g^{-1} *has a left adjoint* $g_!^{Sh}$ *on sheaves of sets,*

(c) *the adjunction maps* $g^{-1}g_* \to id$ *and* $id \to g^{-1}g_!^{Sh}$ *are isomorphisms,*

(d) *the functor* g^{-1} *has a left adjoint* $g_!$ *on abelian sheaves,*

(e) *the adjunction map* $id \to g^{-1}g_!$ *is an isomorphism, and*

(f) *we have* $g^{-1}\mathcal{O}_{\mathcal{X}} = \mathcal{O}_{\mathcal{X}_{flat,fppf}}$ *hence* g *induces a flat morphism of ringed topoi such that* $g^{-1} = g^*$.

Proof. In both cases it is immediate that the functor is fully faithful, continuous, and cocontinuous (see Sites, Definitions 00WV and 00XJ). Hence properties (a), (b), (c) follow from Sites, Lemmas 00XR and 00XT. Parts (d), (e) follow from Modules on Sites, Lemmas 04BG and 077I. Part (f) is immediate. □